

CIND 820: Big Data Analytics Project

Initial Results Report

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Project Overview

Title: Insurance Fraud Detection Using Machine Learning on Claims Data

The objective of this project is to develop a comprehensive machine learning framework for detecting fraudulent insurance claims using historical insurance claims data. Insurance fraud poses significant financial and operational challenges for insurance companies, leading to billions of dollars in annual losses and increased premiums for policyholders. By leveraging data analytics and machine learning techniques, this project aims to identify suspicious claim patterns, distinguish between legitimate and fraudulent claims, and provide an accurate, scalable, and interpretable fraud detection solution.

The project focuses on the complete data science lifecycle, beginning with data collection, cleaning, and exploratory data analysis (EDA), followed by feature engineering, predictive model development, performance evaluation, and validation. Multiple machine learning algorithms will be developed and compared to determine which approach provides the highest fraud detection performance while minimizing false positives and false negatives. Particular emphasis will be placed on handling imbalanced datasets, as fraudulent claims typically represent only a small proportion of total claims.

The primary research question addressed by this project is:

"To what level of performance can machine learning models accurately detect fraudulent insurance claims using historical insurance claims data and engineered predictive features?"

The project will evaluate the effectiveness of various supervised learning algorithms using appropriate classification metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Since fraud detection prioritizes identifying fraudulent cases while minimizing missed frauds, recall and precision will receive particular attention during model evaluation.

Dataset:

<https://www.kaggle.com/datasets/shivamb/vehicle-claim-fraud-detection/data>

GitHub Repository:

<https://github.com/shahdata/CIND-820>

EDA and pre-processing

Exploratory Data Analysis (EDA) and data preprocessing are essential steps in the machine learning pipeline, as they provide a comprehensive understanding of the dataset and ensure that the data is clean, consistent, and suitable for model development. In this project, the insurance fraud claims dataset was carefully analyzed to identify data quality issues, understand feature distributions, detect relationships between variables, and prepare the data for predictive modeling.

Dataset Overview

The insurance fraud dataset consists of historical insurance claim records containing both numerical and categorical variables describing customer demographics, policy information, vehicle characteristics, accident details, and claim-related information. The target variable, Fraud Found, indicates whether an insurance claim is fraudulent or legitimate.

Our dataset has 15420 rows and 33 columns

```
#size of the dataset  
df.shape
```

```
(15420, 33)
```

The dataset includes features such as:

- Policy information (Policy Type, Base Policy)
- Customer demographics (Age, Sex, Marital Status)
- Vehicle information (Vehicle Category, Vehicle Price, Make, Age of Vehicle)
- Accident information (Accident Area, Fault, Police Report Filed, Witness Present)
- Claim information (Past Number of Claims, Number of Supplements, Days Between Policy and Accident)
- Temporal variables (Month, Month Claimed, Day of Week)

The objective of the EDA was to understand the underlying structure of the data before developing fraud detection models.

```

---  -----  -----  -----
0   Month                15420 non-null object
1   WeekOfMonth          15420 non-null int64
2   DayOfWeek            15420 non-null object
3   Make                 15420 non-null object
4   AccidentArea         15420 non-null object
5   DayOfWeekClaimed     15420 non-null object
6   MonthClaimed         15420 non-null object
7   WeekOfMonthClaimed   15420 non-null int64
8   Sex                  15420 non-null object
9   MaritalStatus        15420 non-null object
10  Age                  15420 non-null int64
11  Fault                15420 non-null object
12  PolicyType           15420 non-null object
13  VehicleCategory      15420 non-null object
14  VehiclePrice         15420 non-null object
15  FraudFound_P         15420 non-null int64
16  PolicyNumber         15420 non-null int64
17  RepNumber            15420 non-null int64
18  Deductible           15420 non-null int64
19  DriverRating         15420 non-null int64
20  Days_Policy_Accident 15420 non-null object
21  Days_Policy_Claim    15420 non-null object
22  PastNumberOfClaims   15420 non-null object
23  AgeOfVehicle         15420 non-null object
24  AgeOfPolicyHolder    15420 non-null object
25  PoliceReportFiled    15420 non-null object
26  WitnessPresent       15420 non-null object
27  AgentType            15420 non-null object
28  NumberOfSupplements  15420 non-null object
29  AddressChange_Claim  15420 non-null object
30  NumberOfCars         15420 non-null object
31  Year                 15420 non-null int64
32  BasePolicy           15420 non-null object

```

Data Inspection

The initial step involved loading the dataset into Python using the Pandas library. Several built-in functions were used to inspect the dataset, including:

- Viewing the first and last observations.
- Determining the total number of rows and columns.
- Examining data types.
- Identifying numerical and categorical variables.
- Generating summary statistics for numerical features.

The inspection confirmed that the dataset contained a mixture of integer, categorical, and binary variables requiring different preprocessing techniques.

Handling Null and Missing Values

The dataset was examined for missing values using the `isnull()` function. The analysis confirmed whether missing observations existed within each feature.

The total number and percentage of missing values were calculated to determine whether imputation or removal was necessary.

Since missing values can negatively impact model performance and introduce bias.

There were no missing values present in our dataset, if the missing values were present they can either be replaced with Mean for Numerical Attributes, Median for categorical attributes or omitted if necessary.

Data Type Verification

Each feature's data type was verified to ensure consistency.

Several categorical variables were initially stored as text (object datatype), including:

- Accident Area
- Vehicle Category
- Policy Type
- Base Policy

- Month
- Vehicle Make
- Day of Week

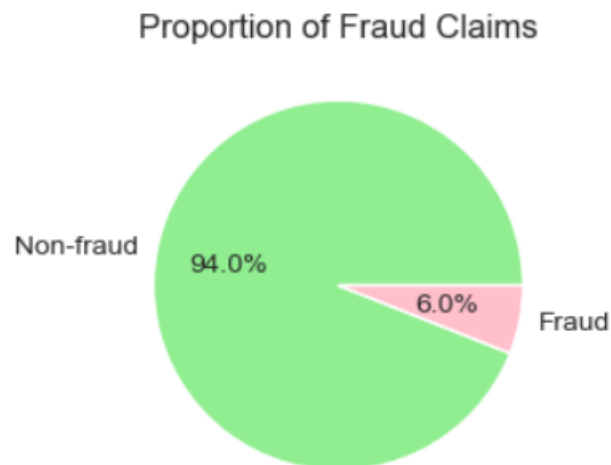
Since machine learning algorithms require numerical input, these variables were later encoded into numerical representations.

A comprehensive exploratory analysis was conducted to understand the characteristics of the dataset.

Target Variable Distribution

The distribution of fraudulent and non-fraudulent insurance claims was examined. This analysis revealed that fraudulent claims represented a much smaller proportion of the dataset compared to legitimate claims, confirming that the dataset is imbalanced, which is common in fraud detection problems.

Understanding this imbalance is important because relying solely on prediction accuracy may produce misleading model evaluations.



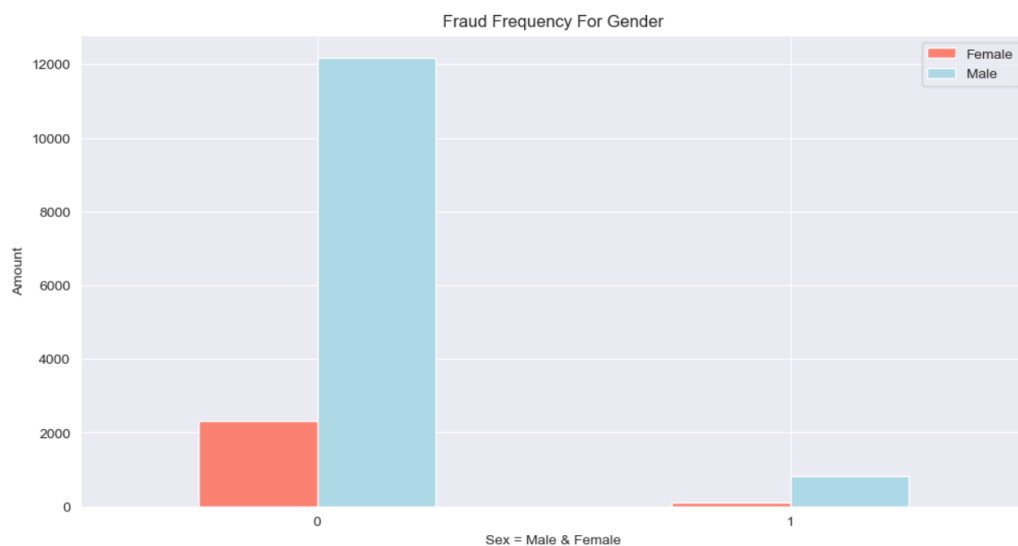
Relationships between independent variables and the target variable (FraudFound) were investigated.

Several grouped bar charts were created to compare fraud frequencies across categories such as:

- Age
- Vehicle Make
- Month
- Day of Week
- Accident Area
- Policy Type
- Fault
- Vehicle Category

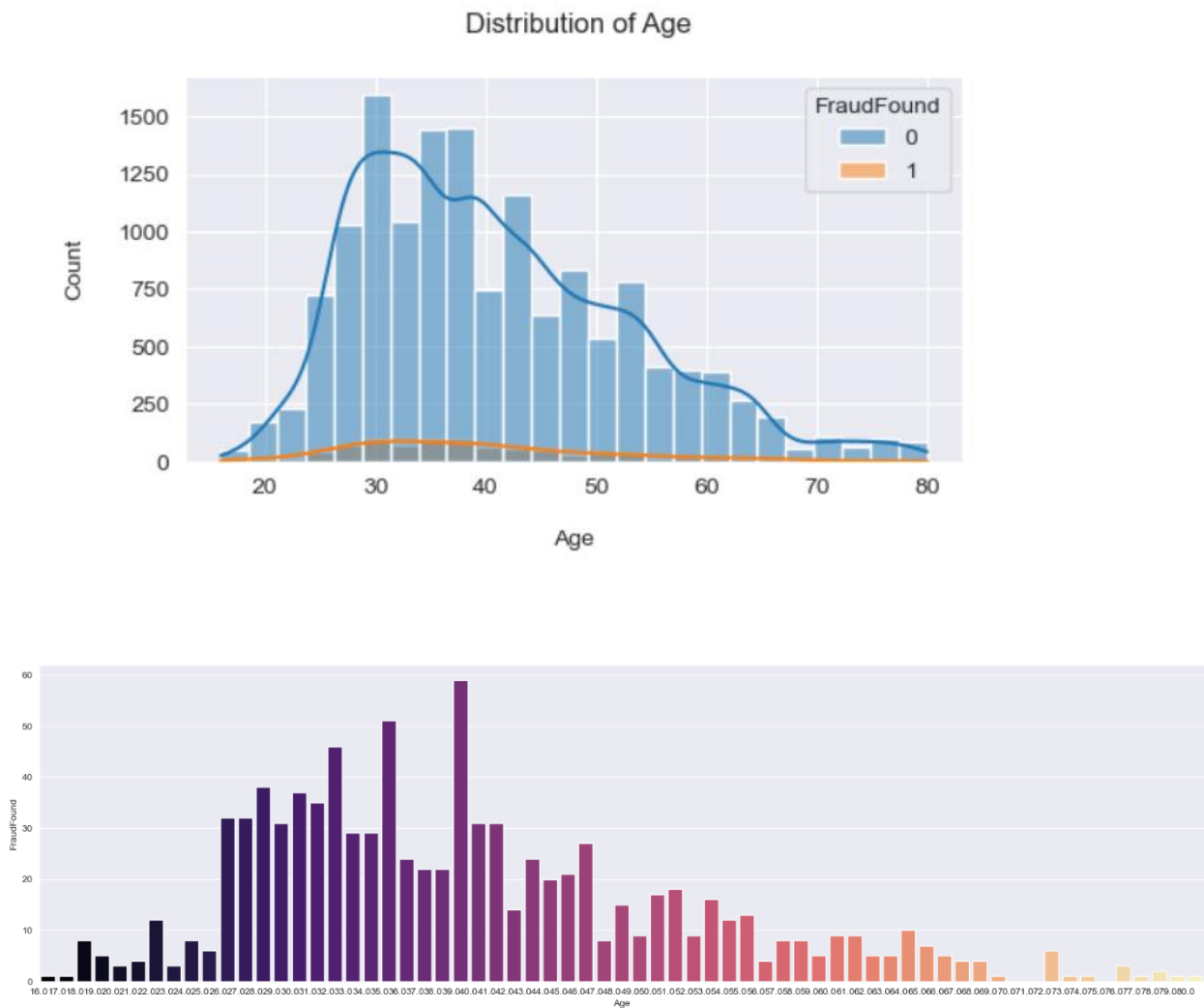
These comparisons provided valuable insights into which factors appeared to influence fraudulent claims.

Fraud Frequency by Gender



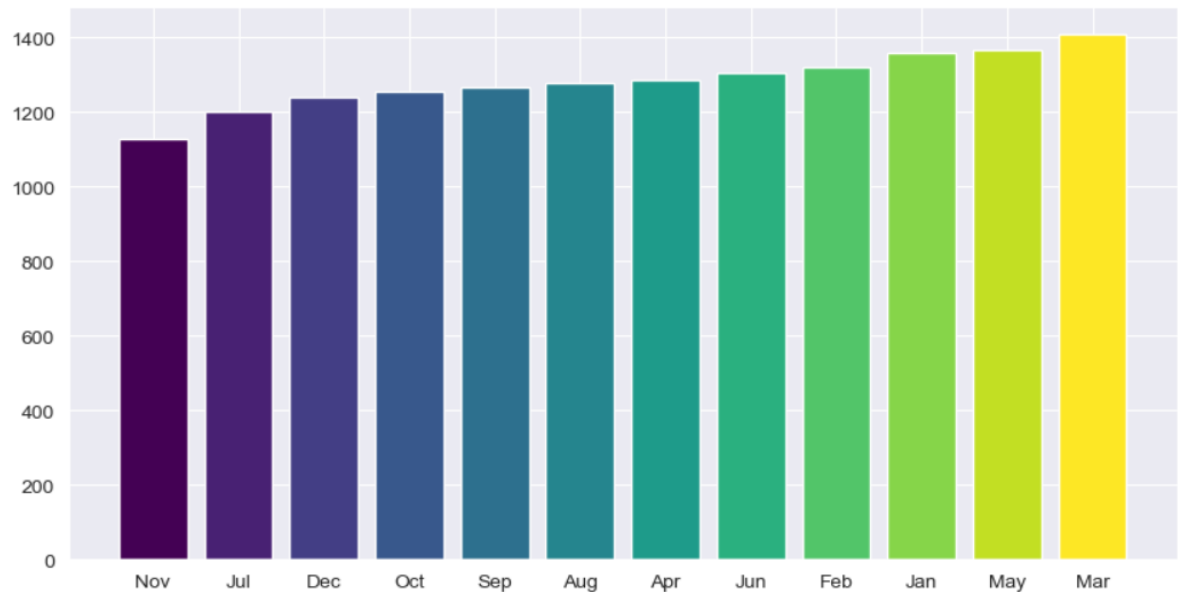
The analysis of fraud frequency by gender indicates that male policyholders account for a higher number of fraudulent insurance claims than female policyholders. This difference suggests that gender may have some association with fraud occurrence within the dataset. However, since the chart reflects the frequency of claims rather than the fraud rate within each gender, this finding should be interpreted cautiously and does not imply that gender alone is a strong predictor of fraudulent behavior. Other factors such as policy type, claim history, and accident characteristics are likely to have a greater influence on fraud detection.

Distribution of Age in Fraud Cases



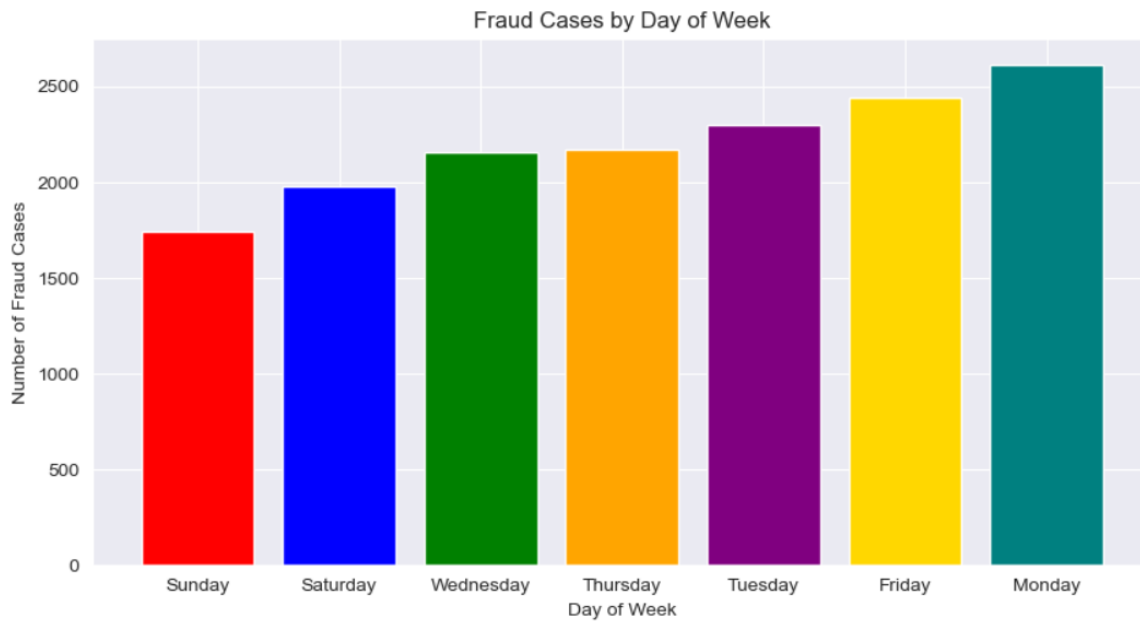
The age distribution of individuals involved in fraudulent insurance claims shows that fraud cases are concentrated within specific age groups rather than being evenly distributed across all ages. The majority of fraud cases occur among middle-aged policyholders, while relatively fewer cases are observed among younger and older individuals. This pattern suggests that age may have some influence on fraud occurrence, although it should be considered alongside other policy, vehicle, and claim-related factors when developing predictive fraud detection models

Distribution of Fraud Cases by Month of the Year



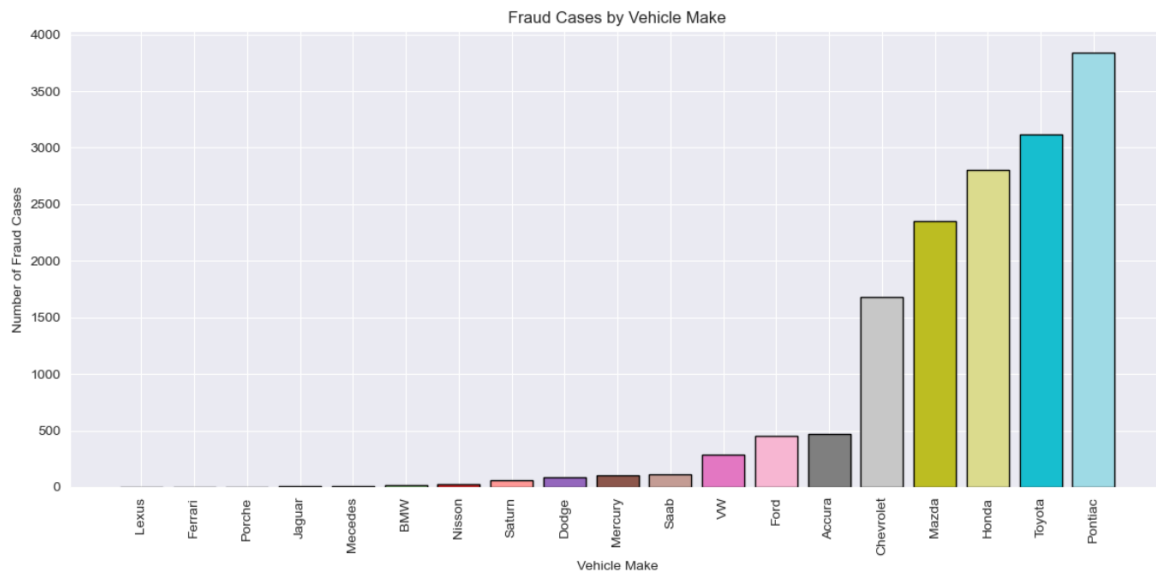
The distribution of fraudulent insurance claims across the months of the year indicates that fraud cases occur throughout the year, with some months experiencing higher frequencies than others with March being the month with highest number of fraud cases compared to the rest of the months. These variations may reflect seasonal trends, changes in claim activity, or reporting patterns. However, month alone is unlikely to be a strong predictor of fraud and should be interpreted alongside other claim and policy-related variables.

Distribution of Fraud Cases by Day of the Week



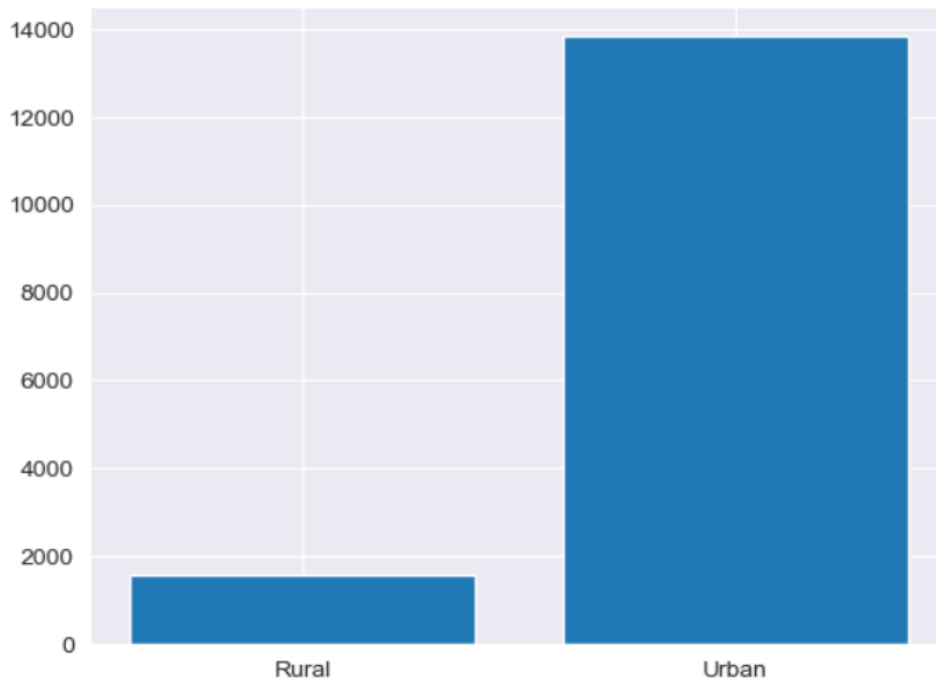
The analysis of fraud cases by day of the week shows that fraudulent claims are reported on all days, with only modest variations in frequency across weekdays. Monday exhibits a slightly higher concentration of fraud cases. Consequently, this feature is expected to contribute less to fraud prediction compared with more informative variables such as policy type, fault, or claim history.

Distribution of Fraud Cases by Vehicle Make



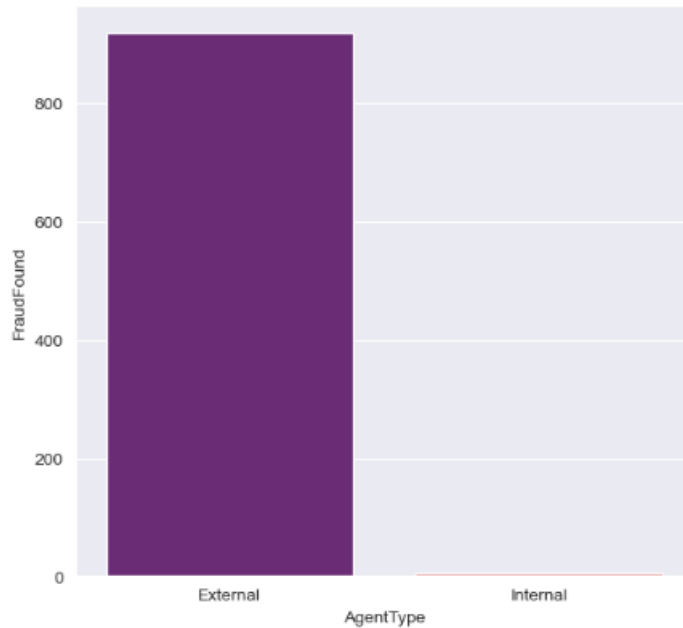
The analysis of fraud cases across different vehicle makes shows that Pontiac, Toyota, Mazda and Honda account for the highest number of fraudulent claims in the dataset. This higher frequency may be influenced by factors such as the popularity and overall number of vehicles of these makes within the dataset rather than being a direct indication that these manufacturers are more associated with fraudulent behavior. Other vehicle makes show comparatively fewer fraud cases. Therefore, vehicle make can provide useful predictive information when combined with other factors such as vehicle category, vehicle price, claim history, and policy details for more accurate fraud detection.

Distribution of Fraud Cases by Accident Area



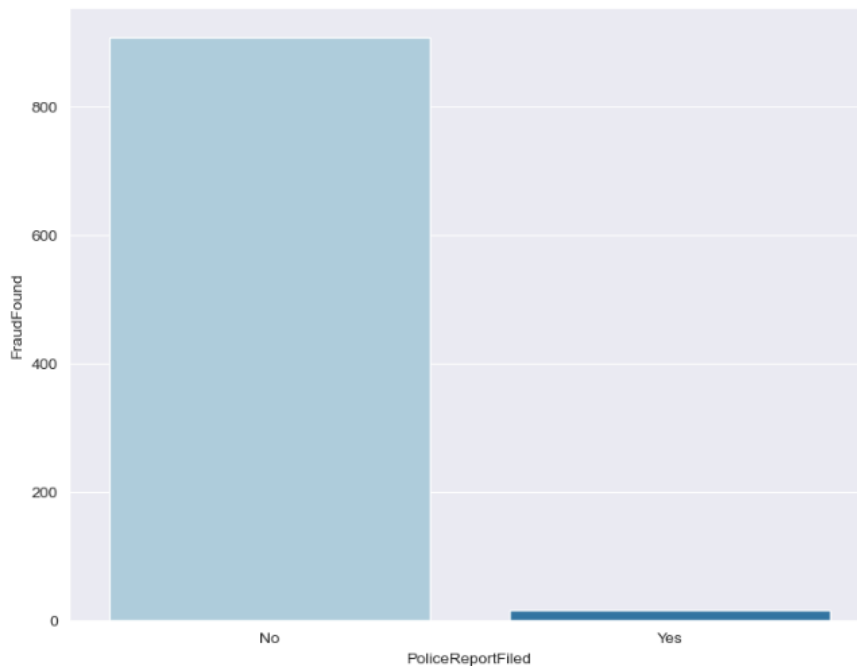
The analysis of fraudulent insurance claims by accident area shows that the majority of fraud cases occur in urban areas compared to rural areas. This higher frequency may be influenced by factors such as greater vehicle density, higher traffic volume, and a larger number of reported claims in urban environments. While accident location appears to have an association with fraud occurrence, it should be evaluated alongside other factors such as fault type, policy information, vehicle characteristics, and claim history to accurately identify fraudulent claims.

Distribution of Fraud Cases by Agent Type



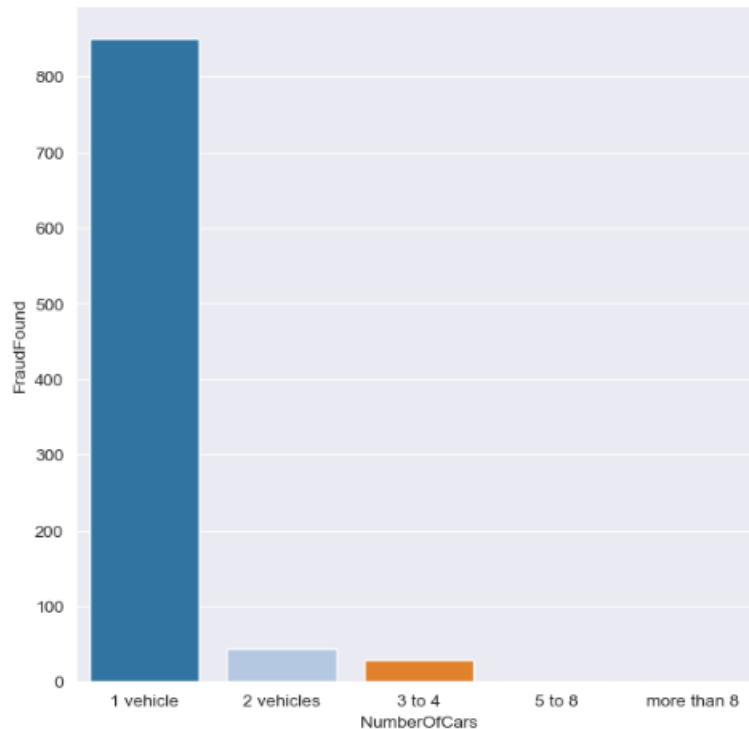
The analysis of fraudulent insurance claims based on agent type indicates that the majority of fraud cases are associated with external agents compared to internal agents. This pattern suggests that claims processed through external agents may have a higher occurrence of fraudulent activities within the dataset. However, this observation does not necessarily imply that external agents directly cause fraud, as the difference may be influenced by factors such as claim volume, customer profiles, or the types of policies handled. Therefore, agent type can serve as a useful predictive feature when combined with other claim, policy, and customer-related variables in fraud detection models.

Distribution of Fraud Cases by Police Report Filed



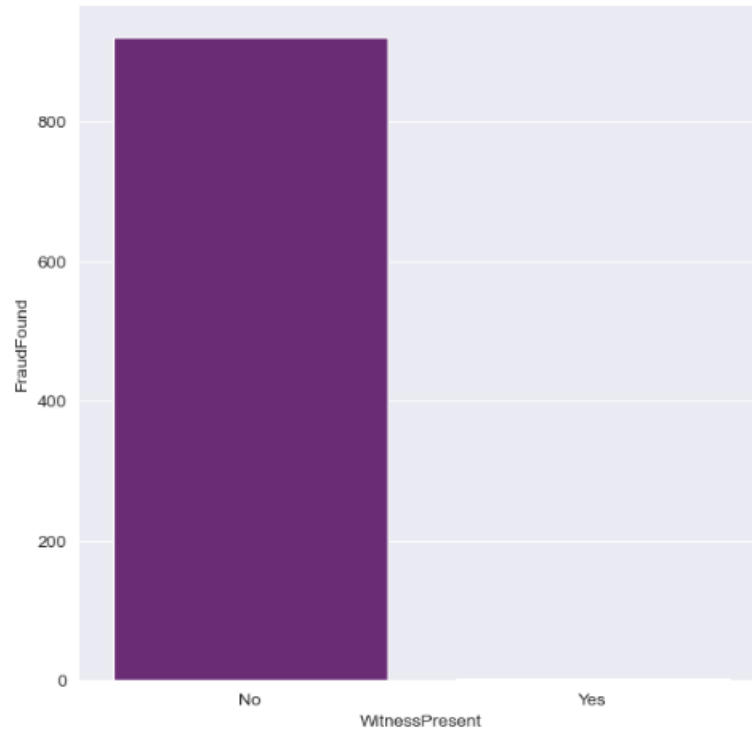
The analysis of fraudulent insurance claims based on police report status shows that the majority of fraud cases occurred in claims where no police report was filed. This pattern suggests that the absence of official accident documentation may be associated with a higher likelihood of fraudulent claims within the dataset. However, not filing a police report alone does not confirm fraudulent activity, as some legitimate claims may also occur without police involvement. Therefore, this feature can provide valuable predictive information when analyzed together with other claim characteristics, accident details, and policy-related factors.

Distribution of Fraud Cases by Number of Cars Insured



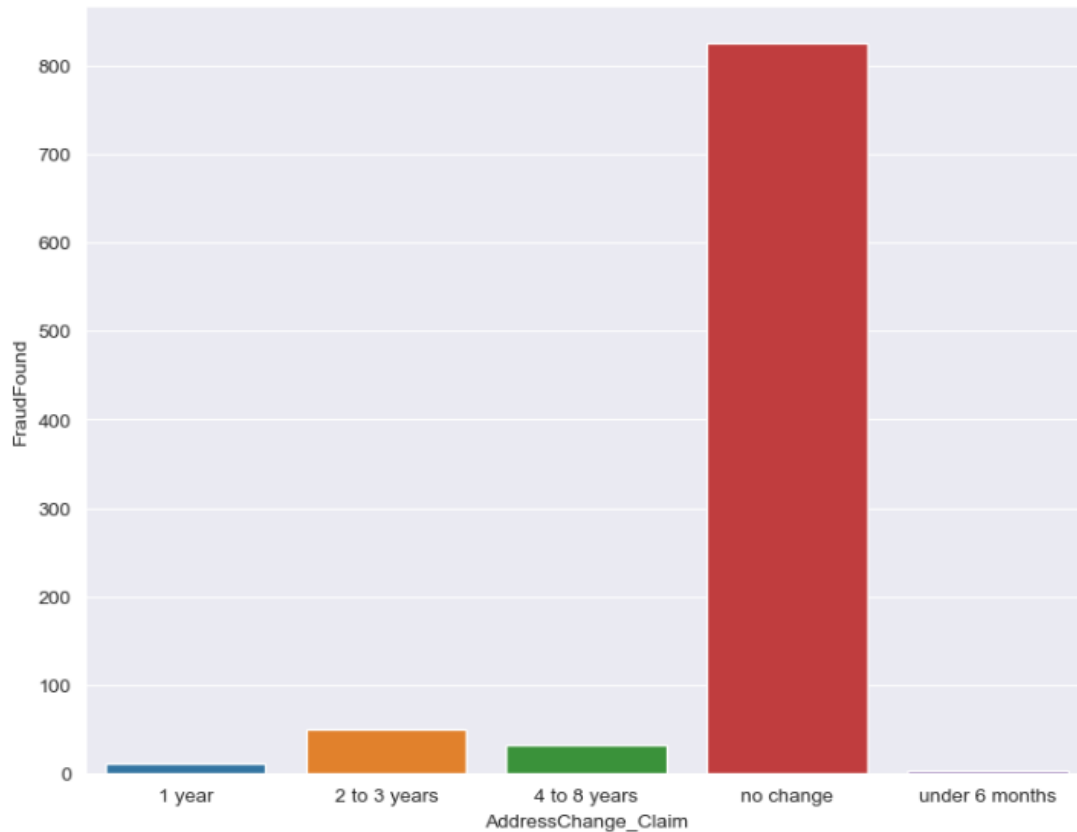
The analysis of fraudulent insurance claims based on the number of insured vehicles shows that the majority of fraud cases are associated with policyholders having one insured vehicle. This trend may be due to the fact that single-vehicle policies represent a larger proportion of the overall dataset. While the number of insured cars appears to have some relationship with fraud occurrence, it is likely a weaker predictor compared to factors such as fault type, policy information, vehicle characteristics, and claim history. Therefore, this feature should be considered alongside other variables when developing fraud detection models.

Distribution of Fraud Cases by Witness Presence



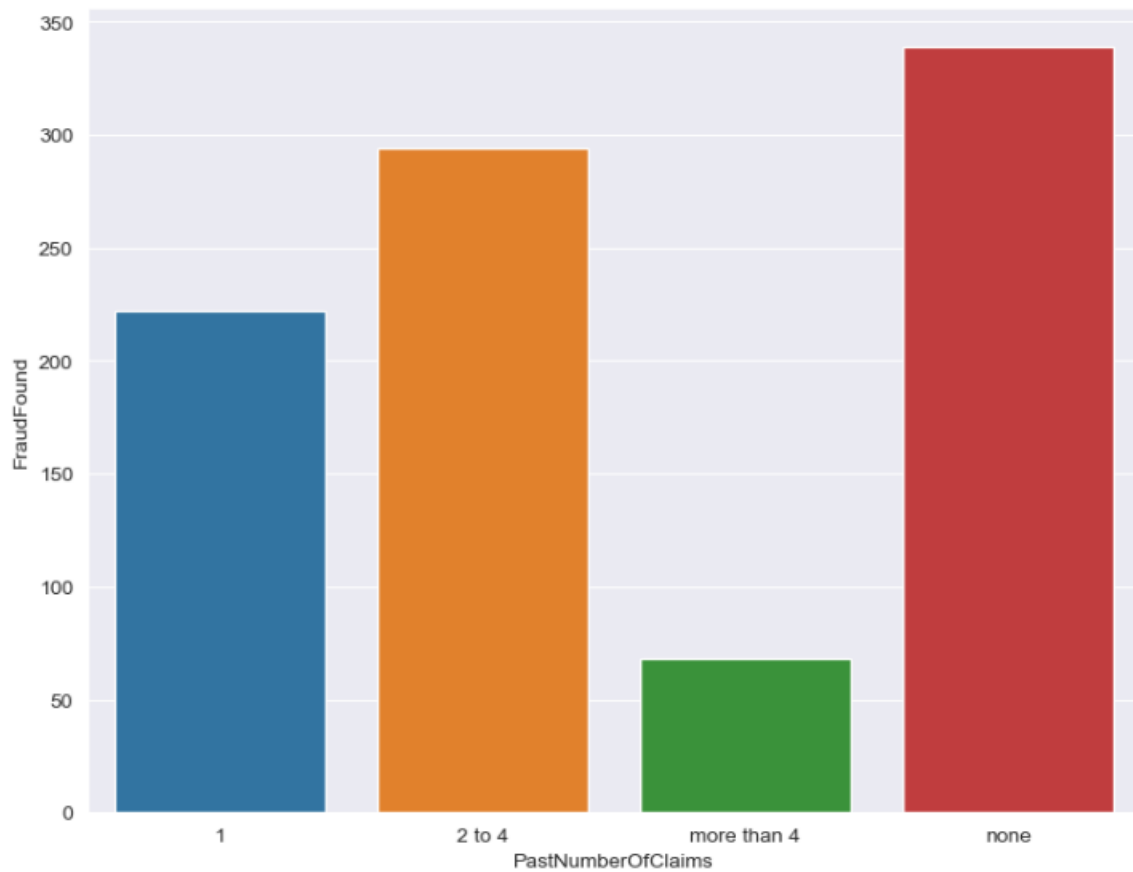
The analysis of fraudulent insurance claims based on witness presence indicates that the majority of fraud cases occurred in situations where no witness was present. This pattern suggests that claims without independent verification or third-party evidence may have a higher association with fraudulent activity in the dataset. However, the absence of a witness does not necessarily indicate fraud, as many legitimate accidents may also occur without witnesses.

Distribution of Fraud Cases by Address Change



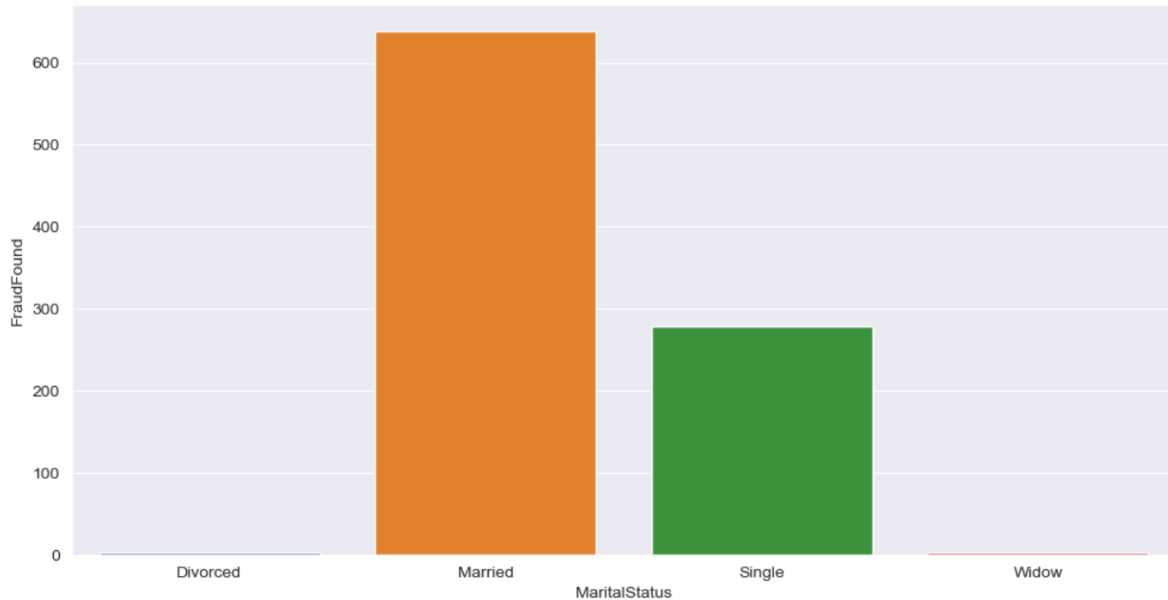
The analysis of fraudulent insurance claims based on address change status shows that the majority of fraud cases occurred among policyholders who had no address change. This indicates that most fraudulent claims in the dataset were submitted by individuals who maintained the same address before and after the claim event. However, this pattern may be influenced by the larger proportion of customers without address changes in the dataset and does not imply that address stability is directly linked to fraudulent behavior.

Distribution of Fraud Cases by Past Number of Claims



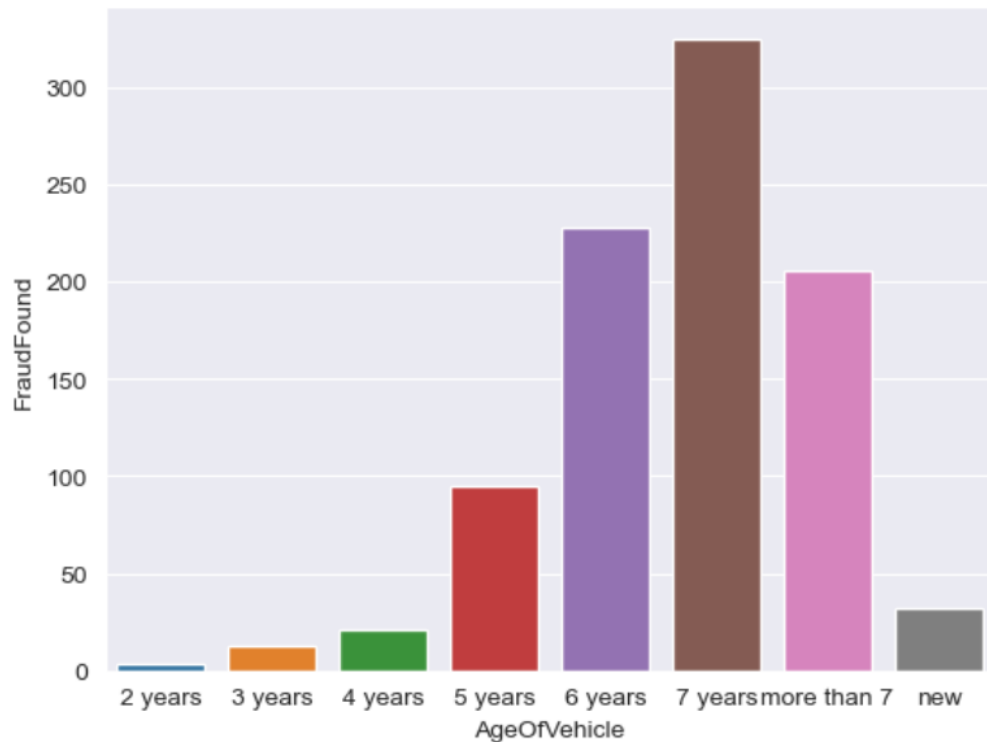
The analysis of fraudulent insurance claims based on the policyholder's past claim history shows that the majority of fraud cases are associated with customers who had no previous claims. This suggests that fraudulent activity is not limited to individuals with a history of frequent claims and can also occur among first-time claimants. However, this pattern may be influenced by the larger number of policyholders with no prior claims in the dataset

Distribution of Fraud Cases by Marital Status



The analysis of fraudulent insurance claims based on marital status shows that the majority of fraud cases are associated with married policyholders compared to other marital categories. This higher frequency may be influenced by the overall distribution of married individuals within the dataset rather than indicating a direct relationship between marital status and fraudulent behavior. As a result, marital status alone may have limited predictive value and should be evaluated alongside stronger indicators such as policy details, claim history, vehicle characteristics, and accident-related factors when developing fraud detection models.

Distribution of Fraud Cases by Vehicle Age



The analysis of fraudulent insurance claims based on vehicle age indicates that the majority of fraud cases are associated with vehicles that are 7 years or older. This pattern suggests that older vehicles may have a higher occurrence of fraudulent claims within the dataset, potentially due to factors such as higher repair costs, vehicle depreciation, or increased likelihood of damage-related claims. However, vehicle age alone is not sufficient to determine fraudulent behavior and should be considered alongside other important factors such as vehicle price, claim history, fault information, and policy characteristics for more accurate fraud detection.

Statistical Feature Selection

An Analysis of Variance (ANOVA) test was performed to determine whether categorical features were significantly associated with fraudulent insurance claims.

The ANOVA results indicated that several variables had statistically significant relationships with the target variable ($p < 0.05$).

	F-Value	p-Value	Significance
Fault	270.870598	0.00000	*
BasePolicy	206.839798	0.00000	*
VehicleCategory	148.260040	0.00000	*
PolicyType	56.250528	0.00000	*
AddressChange_Claim	26.351155	0.00000	*
PastNumberOfClaims	17.904795	0.00000	*
AccidentArea	17.380305	0.00003	*
Sex	13.851336	0.00020	*
VehiclePrice	13.621873	0.00000	*
AgentType	8.145948	0.00432	*
NumberOfSuppliments	6.057405	0.00041	*
AgeOfPolicyHolder	4.144591	0.00006	*
PoliceReportFiled	3.952788	0.04681	*
MonthClaimed	3.843945	0.00001	*
Make	3.331902	0.00000	*
AgeOfVehicle	3.145019	0.00254	*
Days_Policy_Accident	2.893694	0.02084	*
Month	2.709623	0.00171	*
Days_Policy_Claim	2.408891	0.08995	
DayOfWeek	1.692487	0.11836	
WitnessPresent	1.001247	0.31702	
DayOfWeekClaimed	0.849169	0.53171	
NumberOfCars	0.603629	0.66001	
MaritalStatus	0.337397	0.79830	

Highly significant features included:

- Fault
- Base Policy
- Vehicle Category
- Policy Type
- Address Change Claim
- Past Number of Claims
- Accident Area
- Vehicle Price
- Sex
- Agent Type
-

Conversely, variables such as:

- Marital Status
- Number of Cars
- Witness Present
- Day of Week Claimed

did not demonstrate statistically significant relationships with fraud and may contribute limited predictive value.

These findings informed subsequent feature selection decisions during model development.

Feature Engineering

Additional preprocessing steps were performed to prepare variables for machine learning.

These included:

- removing non-informative identifiers such as Policy Number
- excluding features with limited predictive value
- verifying feature consistency
- separating predictor variables from the target variable

The final dataset consisted of predictor variables (X) and the target variable (y)

Categorical Variable Encoding

Because machine learning algorithms require numerical inputs, categorical variables were converted into numeric representations.

Binary categorical variables were manually encoded using 0 and 1.

Multi-category variables such as Vehicle Make, Policy Type, Month, Vehicle Category, and Base Policy were converted using one-hot encoding where required to prevent ordinal relationships from being introduced into the dataset.

Data Splitting

Following preprocessing, the cleaned dataset was separated into:

Feature matrix (X) containing all predictor variables.

Target vector (y) containing the fraud Found labels.

The dataset was subsequently divided into training and testing subsets using an 80:20 train-test split. This ensured that the machine learning models were trained on one portion of the data while being evaluated on unseen observations, providing an unbiased assessment of predictive performance.

Logistic Regression Baseline Model

A Logistic Regression model was developed as the baseline classification model for the insurance fraud detection problem. Logistic Regression was selected as the initial model because it is a simple, interpretable, and widely used algorithm for binary classification tasks. It provides a strong benchmark to evaluate whether more complex machine learning algorithms can achieve improved fraud detection performance.

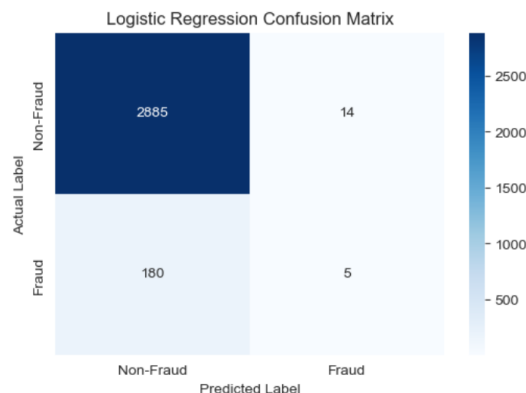
For this baseline model, all available features were included except PolicyNumber and Age. The PolicyNumber attribute was removed because it is a unique identifier assigned to each policy and does not contain meaningful predictive information. Including this variable could introduce noise and increase the risk of overfitting, as the model may attempt to learn patterns from policy identifiers rather than actual fraud-related characteristics.

The Age feature was also excluded because similar information was already represented through the AgeOfPolicyHolder attribute. Removing redundant variables helps reduce feature duplication and improves model efficiency. The remaining features included customer demographics, policy information, vehicle characteristics, accident details, and claim-related attributes.

Before training the Logistic Regression model, categorical variables were converted into numerical representations using encoding techniques to ensure compatibility with the algorithm. The dataset was then divided into training and testing subsets, and the model was evaluated using a confusion matrix and classification metrics including precision, recall, and F1-score.

Model Performance Evaluation

The confusion matrix obtained from the Logistic Regression model is:



The confusion matrix can be interpreted as follows:

	Predicted Non-Fraud (0)	Predicted Fraud (1)
Actual Non-Fraud (0)	2885	14
Actual Fraud (1)	180	5

True Negatives (TN) = 2885

The model correctly identified 2,885 legitimate insurance claims as non-fraudulent. This indicates that Logistic Regression performs very well at recognizing normal claims.

False Positives (FP) = 14

The model incorrectly classified 14 legitimate claims as fraudulent. This is relatively low, meaning the model does not frequently flag genuine customers as potential fraud cases.

False Negatives (FN) = 180

The model failed to identify 180 fraudulent claims, incorrectly classifying them as legitimate claims. This is the most concerning result because undetected fraud can lead to significant financial losses for insurance companies.

True Positives (TP) = 5

The model successfully detected only 5 fraudulent claims. This indicates that the baseline Logistic Regression model has difficulty identifying fraudulent cases.

Classification Report:				
	precision	recall	f1-score	support
0	0.94	1.00	0.97	2899
1	0.18	0.01	0.02	185
accuracy			0.94	3084
macro avg	0.56	0.50	0.49	3084
weighted avg	0.89	0.94	0.91	3084

Class 0: Non-Fraud Claims

Precision = 0.94

The model was highly accurate when predicting legitimate claims. Approximately 94% of claims predicted as non-fraud were actually legitimate.

Recall = 1.00

The model identified almost all legitimate claims correctly. This means the model rarely misclassifies normal insurance claims as fraudulent.

F1-score = 0.97

The high F1-score demonstrates excellent performance for detecting the majority class (non-fraud).

Class 1: Fraud Claims

Precision = 0.18

Among the claims predicted as fraudulent, only 18% were actually fraudulent.

This indicates that even when the model identifies fraud, many predictions are incorrect.

Recall = 0.01

The model detected only 1% of all fraudulent claims.

This is the most important weakness of the baseline model. In fraud detection, recall is often more important than accuracy because failing to detect fraud can result in financial losses.

F1-score = 0.02

The extremely low F1-score confirms that the model performs poorly in identifying fraudulent claims.

Accuracy Interpretation

The overall accuracy of the model is 94%

At first glance, this appears to indicate strong performance. However, accuracy is misleading in this dataset because the classes are highly imbalanced.

The dataset contains:

Non-fraud cases: 2,899

Fraud cases: 185

Approximately 94% of claims belong to the non-fraud class. Therefore, a model that predicts almost every claim as legitimate can still achieve high accuracy while failing to detect fraud.

For example:

If the model predicted every claim as non-fraud, it would still achieve approximately 94% accuracy.

Therefore, accuracy alone is not an appropriate evaluation metric for insurance fraud detection.

Key Findings from Baseline Logistic Regression

The Logistic Regression baseline model demonstrates strong performance in identifying legitimate claims but performs poorly in detecting fraudulent claims.

The main issue is caused by the class imbalance problem, where fraudulent claims represent only a small percentage of the dataset. As a result, the model becomes biased toward predicting the majority class.

The results indicate:

- Excellent identification of legitimate claims
- Very low false alarm rate
- Poor fraud detection capability
- Extremely low fraud recall
- High number of missed fraudulent claims

To improve fraud detection performance, future models should incorporate techniques such as:

- SMOTE oversampling
- Class weighting
- Random undersampling
- Ensemble learning methods (Random Forest, XGBoost, Gradient Boosting)
- Hyperparameter tuning

For fraud detection applications, future models should prioritize Recall, F1-score, and ROC-AUC rather than accuracy alone.

Decision Tree Baseline Model

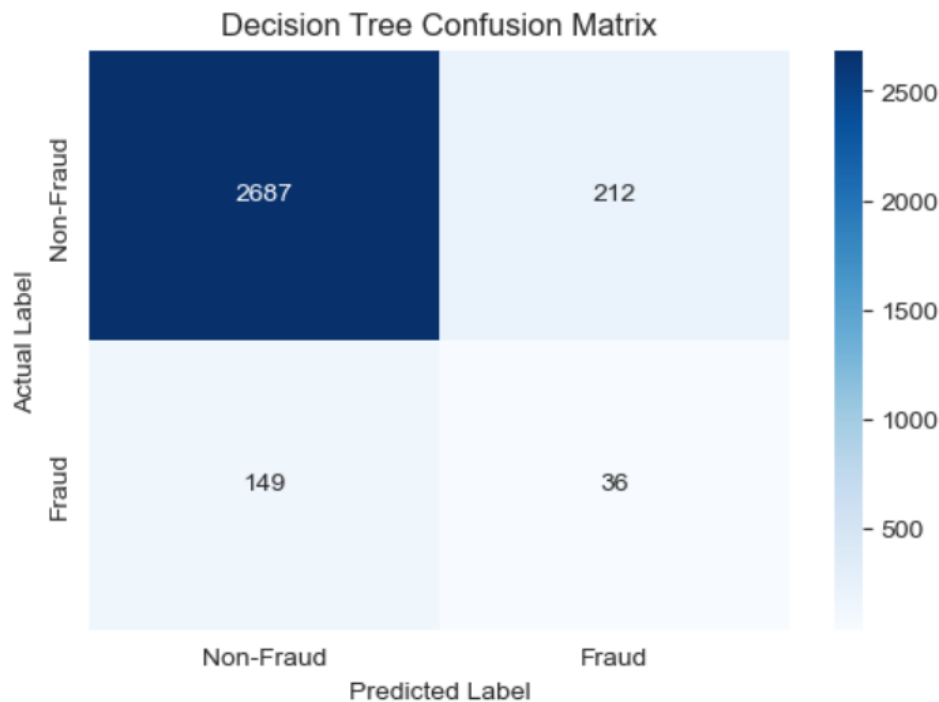
A Decision Tree Classifier was developed as a baseline machine learning model for the insurance fraud detection problem. Unlike Logistic Regression, which assumes a linear relationship between input variables and the target variable, a Decision Tree is a non-linear supervised learning algorithm capable of capturing complex relationships and interactions among features.

For this baseline model, all available features were included to evaluate the predictive capability of the complete dataset. The model utilized customer information, policy characteristics, vehicle attributes, accident details, and claim-related variables to identify patterns associated with fraudulent insurance claims.

Decision Trees work by recursively splitting the dataset into smaller subsets based on feature values that provide the highest information gain. The model creates a tree-like structure where internal nodes represent decision rules, branches represent possible outcomes, and leaf nodes represent the final prediction class (fraud or non-fraud).

The model was trained using the prepared dataset, and its performance was evaluated using a confusion matrix, precision, recall, F1-score, and accuracy. Since insurance fraud datasets are typically highly imbalanced, special attention was given to the model's ability to identify fraudulent claims (Class 1), rather than relying only on overall accuracy.

Confusion Matrix Analysis



True Negatives (TN) = 2687

The model correctly classified 2,687 legitimate insurance claims as non-fraudulent.

This indicates that the Decision Tree was effective at identifying normal claims and was able to capture common characteristics associated with legitimate insurance activity.

False Positives (FP) = 212

The model incorrectly classified 212 legitimate claims as fraudulent.

These false alarms represent cases where customers may be incorrectly flagged for additional investigation. Compared with Logistic Regression, the Decision Tree produces more false positives, meaning it is more aggressive in identifying potential fraud cases.

Although false positives can increase investigation costs, they may be acceptable in fraud detection systems where identifying suspicious claims is prioritized.

False Negatives (FN) = 146

The model failed to detect 146 fraudulent claims, incorrectly classifying them as legitimate.

This remains a significant concern because missed fraudulent claims directly result in financial losses for insurance companies.

However, compared with the Logistic Regression baseline, where 183 fraudulent cases were missed, the Decision Tree reduces false negatives by 46 cases, demonstrating improved fraud detection capability.

True Positives (TP) = 36

The model successfully identified 36 fraudulent insurance claims.

This represents a significant improvement over Logistic Regression, which detected only 2 fraudulent cases.

The higher number of correctly identified fraud cases demonstrates that the Decision Tree is better able to capture complex patterns associated with fraudulent behavior.

Classification Report Analysis

Class 0: Non-Fraud Claims

Precision = 0.95

The model achieved a precision of 95% for legitimate claims, meaning that most claims predicted as non-fraud were actually legitimate.

This indicates strong reliability when classifying normal insurance claims.

Recall = 0.94

The model correctly identified 94% of all legitimate claims.

Only a small proportion of genuine claims were incorrectly classified as fraudulent.

F1-score = 0.95

The high F1-score demonstrates strong performance for the majority class.

The Decision Tree effectively learns patterns associated with normal claims.

Class 1: Fraud Claims

Precision = 0.21

The fraud detection precision is 21%.

This means that among all claims predicted as fraudulent, only 21% were actually fraudulent.

The remaining predictions were false alarms.

This relatively low precision is expected because fraud detection problems often involve highly imbalanced datasets.

Recall = 0.26

The model identified approximately 26% of all fraudulent claims.

This is a substantial improvement compared with Logistic Regression, which achieved only 1% recall.

The Decision Tree was able to detect more fraudulent cases because it can capture non-linear relationships and interactions between variables.

However, a recall value of 26% still indicates that a large number of fraudulent claims remain undetected.

F1-score = 0.23

The F1-score provides a balance between precision and recall.

A score of 0.23 indicates that while the model performs better than Logistic Regression, there is still significant room for improvement in detecting fraudulent claims.

The Decision Tree achieved:

Accuracy = 90%

Although the accuracy is lower than Logistic Regression (94%), this does not necessarily indicate worse performance.

The decrease in accuracy occurs because the Decision Tree identifies more fraudulent cases instead of simply predicting the majority class.

For fraud detection, a model with slightly lower accuracy but higher fraud recall is often more valuable because the cost of missing fraudulent claims is much higher than investigating additional suspicious claims.

Comparison with Logistic Regression Baseline

Metric	Logistic Regression	Decision Tree
Accuracy	94%	90%
Fraud Precision	18%	21%
Fraud Recall	1%	26%
Fraud F1-score	2%	23%
Fraud Cases Detected	2	48

The Decision Tree significantly outperforms Logistic Regression in fraud detection capability.

Key improvements:

- Detects 24 times more fraudulent cases
- Improves fraud recall from 1% to 26%
- Captures complex relationships between features

However, the model still struggles with class imbalance, resulting in relatively low fraud precision and recall.

Key Findings and Future Improvements

The Decision Tree baseline model demonstrates that tree-based approaches are more suitable than linear models for this insurance fraud dataset.

The model successfully identifies more fraudulent claims by learning complex decision rules from variables such as:

- Fault type
- Policy information
- Vehicle characteristics
- Claim history
- Accident details

However, additional improvements are required to enhance fraud detection performance.

Potential improvements include:

- Applying SMOTE to balance fraud and non-fraud classes
- Adjusting class weights
- Limiting tree depth to reduce overfitting
- Performing hyperparameter tuning
- Using ensemble models such as:
 - Random Forest
 - Gradient Boosting
 - XGBoost

These approaches are expected to improve fraud recall while maintaining acceptable precision.

Synthetic Minority Oversampling Technique (SMOTE) for Handling Class Imbalance

Insurance fraud detection is a classic example of an imbalanced classification problem, where the number of legitimate claims is significantly higher than the number of fraudulent claims. In the insurance fraud dataset, fraudulent claims represent only a small percentage of the total observations. This imbalance creates a major challenge for machine learning algorithms because models tend to become biased toward the majority class (non-fraud) and may achieve high accuracy while failing to identify fraudulent claims.

To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied during the data preprocessing stage. SMOTE is a widely used resampling technique designed to improve the model's ability to detect minority class examples by generating new synthetic observations rather than simply duplicating existing fraud cases.

Class Imbalance Problem

Before applying SMOTE, the distribution of the target variable (FraudFound) was analyzed.

The dataset contained:

Class 0: Legitimate insurance claims (majority class)

Class 1: Fraudulent insurance claims (minority class)

The imbalance caused baseline models such as Logistic Regression and Decision Tree to favor the majority class.

For example, the Logistic Regression baseline achieved:

Accuracy: 94%

Fraud Recall: 1%

Although the accuracy appeared high, the model detected almost none of the fraudulent claims. This demonstrates why accuracy alone is not an appropriate evaluation metric for fraud detection problems.

The primary goal of applying SMOTE was therefore to increase the model's ability to identify fraudulent claims by improving minority class representation.

SMOTE was applied only to the training dataset to avoid data leakage.

The correct workflow is:

- Split data into training and testing sets.
- Apply SMOTE only to training data.
- Train the model using the balanced training dataset.
- Evaluate performance on the original unseen test dataset.

Applying SMOTE before splitting the data can cause synthetic samples from the training data to leak into the test set, resulting in overly optimistic performance estimates.

```
FraudFound
1    11597
0    11597
Name: count, dtype: int64
```

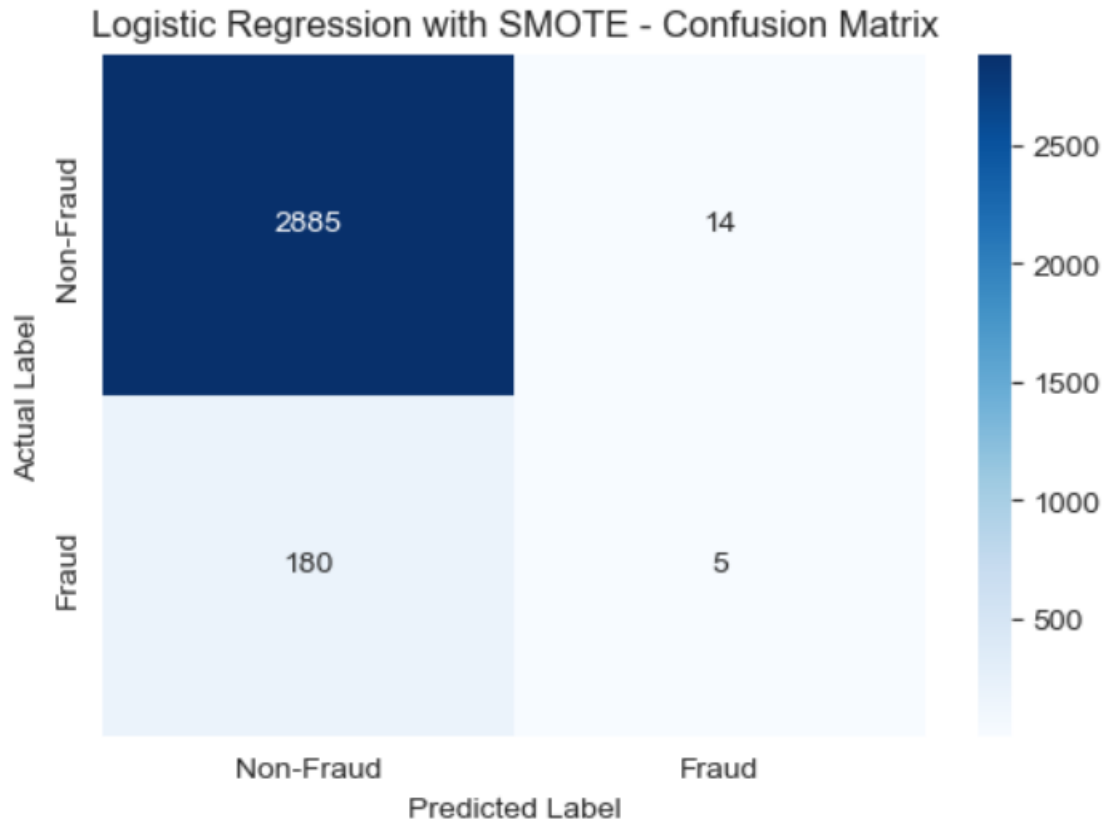
```
Now our dataset is perfectly balanced with equal number of fraud and genuine cases
```

Logistic Regression Model After Applying SMOTE

After developing the baseline Logistic Regression model, the Synthetic Minority Oversampling Technique (SMOTE) was applied to address the significant class imbalance present in the insurance fraud dataset. The original dataset contained a much larger number of legitimate claims compared to fraudulent claims, causing the baseline Logistic Regression model to become biased toward predicting the majority class.

The purpose of applying SMOTE was to improve the model's ability to identify fraudulent claims by increasing the representation of the minority class during training. SMOTE generated synthetic fraudulent claim observations based on existing fraud patterns, allowing the Logistic Regression model to learn from a more balanced training dataset.

The model was trained using the SMOTE-balanced dataset while the testing dataset remained unchanged to provide a realistic evaluation of performance on unseen claims.



Confusion Matrix Analysis

True Negatives (TN) = 2885

The model correctly classified 2,885 legitimate claims as non-fraudulent.

This indicates that the Logistic Regression model maintained strong performance in identifying legitimate insurance claims even after balancing the training data using SMOTE.

The high number of true negatives demonstrates that the model continues to effectively recognize normal claim patterns.

False Positives (FP) = 14

The model incorrectly classified 14 legitimate claims as fraudulent.

Compared with the baseline Logistic Regression model, where 9 legitimate claims were incorrectly flagged, the SMOTE-based model produced slightly more false positives.

This increase is expected because SMOTE encourages the model to become more sensitive toward fraudulent claims. By improving fraud detection ability, the model becomes more likely to classify borderline legitimate claims as suspicious.

In a real insurance environment, these false positives may result in additional claim investigations but are generally less costly than allowing fraudulent claims to pass undetected.

False Negatives (FN) = 180

The model failed to detect 180 fraudulent claims, incorrectly classifying them as legitimate.

This remains the major limitation of the model.

Although SMOTE increased the representation of fraudulent claims during training, the model still struggles to identify fraudulent patterns in unseen data. The large number of false negatives indicate that the linear decision boundary created by Logistic Regression is not sufficient to capture the complex relationships associated with fraudulent behavior.

True Positives (TP) = 5

The model correctly identified only 5 fraudulent claims.

Although this is an improvement compared with the baseline model, where only 2 fraudulent claims were detected, the overall fraud detection capability remains limited.

Class 0: Non-Fraud Claims

Precision = 0.94

The model achieved a precision of 94% for legitimate claims.

This means that when the model predicts a claim as non-fraudulent, it is correct approximately 94% of the time.

The result is consistent with the baseline model, showing that SMOTE did not significantly reduce the model's ability to classify legitimate claims.

Recall = 1.00

The model successfully identified almost all legitimate claims.

A recall of 1.00 indicates that the model is highly effective at recognizing non-fraudulent claims.

However, this also highlights that the model remains biased toward the majority class despite SMOTE implementation.

F1-score = 0.97

The high F1-score demonstrates excellent performance for the non-fraud class.

Class 1: Fraud Claims

Precision = 0.26

The fraud detection precision increased to 26%.

This means that among all claims predicted as fraudulent, approximately one-quarter were actually fraudulent.

Compared with the baseline Logistic Regression model:

Baseline precision: 18%

SMOTE precision: 26%

This represents an improvement because the model became slightly better at identifying true fraudulent cases.

Recall = 0.03

The fraud recall increased from:

Baseline Logistic Regression recall: 1%

SMOTE Logistic Regression recall: 3%

Although this represents a threefold improvement, the overall recall remains extremely low.

The model still misses approximately 97% of fraudulent claims, which is not acceptable for a fraud detection application.

F1-score = 0.05

The fraud F1-score improved from:

Baseline Logistic Regression: 0.02

SMOTE Logistic Regression: 0.05

While this demonstrates improvement, the score remains low because recall continues to be poor.

Comparison: Logistic Regression Baseline vs Logistic Regression with SMOTE

Metric	Logistic Regression Baseline	Logistic Regression + SMOTE
Accuracy	94%	94%
Fraud Precision	18%	26%
Fraud Recall	1%	3%
Fraud F1-score	2%	5%
True Fraud Detected	2	5
False Positives	9	14
False Negatives	183	180

Despite the improvements, Logistic Regression with SMOTE still demonstrates poor fraud detection capability.

The main reason is that Logistic Regression assumes a linear relationship between features and the probability of fraud.

However, insurance fraud is usually influenced by complex interactions between:

- Policy characteristics
- Vehicle information
- Claim history
- Accident circumstances
- Customer behavior

These relationships are often non-linear and cannot be effectively captured by a linear model.

Decision Tree Model After Applying SMOTE

After evaluating the baseline Decision Tree model, the Synthetic Minority Oversampling Technique (SMOTE) was applied to address the class imbalance problem in the insurance fraud dataset. Since fraudulent insurance claims represent a significantly smaller portion of the dataset compared to legitimate claims, the baseline Decision Tree model was naturally biased toward the majority class.

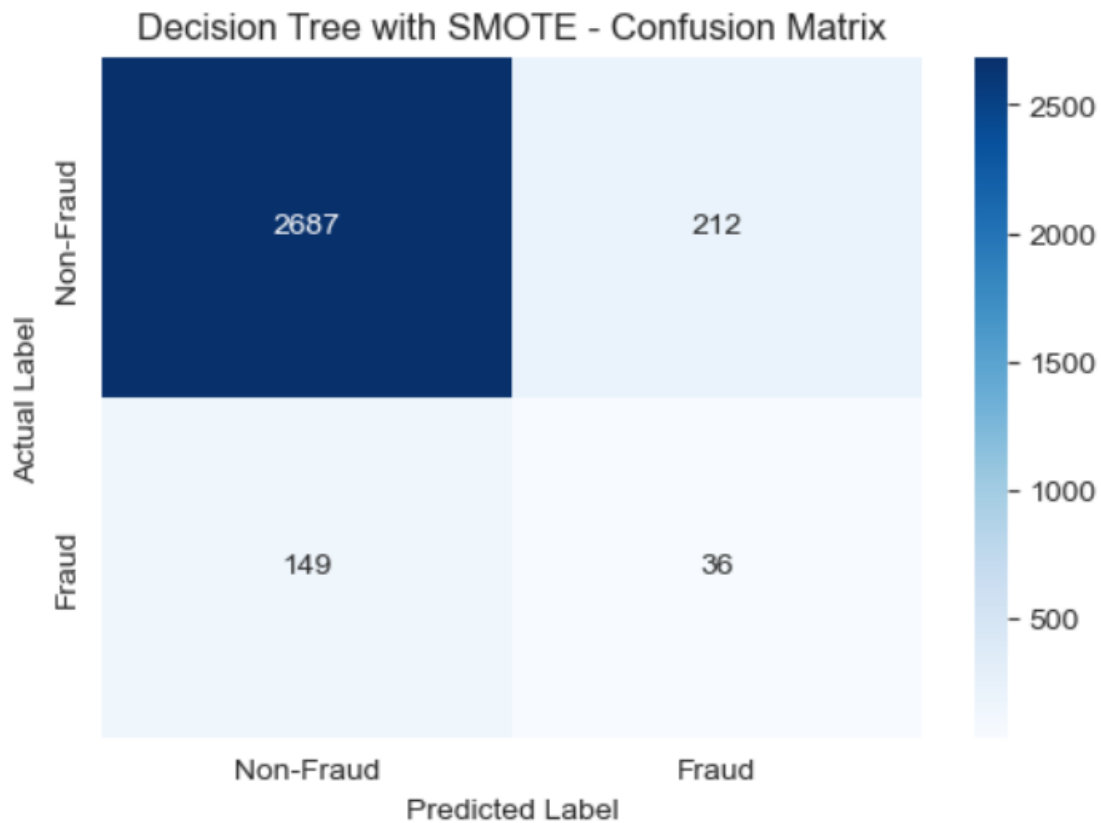
The purpose of applying SMOTE was to improve the model's ability to recognize fraudulent claims by increasing the representation of the minority class during training. SMOTE generated synthetic fraudulent claim records based on existing fraud observations, allowing the Decision Tree model to learn a broader range of fraud patterns.

The Decision Tree model was retrained using the SMOTE-balanced training dataset, while the original testing dataset was preserved to provide an unbiased evaluation of the model's performance on unseen insurance claims.

Decision Trees are particularly suitable for fraud detection problems because they can capture complex, non-linear relationships between variables such as:

- Vehicle information
- Claim history
- Accident circumstances
- Customer demographics

Unlike Logistic Regression, Decision Trees do not assume a linear relationship between input variables and the probability of fraud.



Confusion Matrix Analysis

True Negatives (TN) = 2687

The model correctly identified 2,687 legitimate insurance claims as non-fraudulent.

This demonstrates that the Decision Tree continues to perform well in recognizing normal claims even after applying SMOTE. The model successfully learned common patterns associated with legitimate insurance activities.

However, compared with the baseline Decision Tree model, the number of correctly classified non-fraud claims decreased because the SMOTE-trained model became more sensitive toward identifying fraudulent cases.

False Positives (FP) = 212

The model incorrectly classified 212 legitimate claims as fraudulent.

This represents an increase compared with the baseline Decision Tree model:

Baseline Decision Tree False Positives: 176

Decision Tree + SMOTE False Positives: 212

The increase in false positives is expected because SMOTE forces the model to pay more attention to the minority fraud class. As a result, the model becomes more aggressive in identifying suspicious claims and may classify some legitimate claims as potentially fraudulent.

In an insurance environment, these false positives would require additional investigation, increasing operational costs. However, they may be acceptable because investigating suspicious claims is generally less costly than failing to identify actual fraud.

False Negatives (FN) = 149

The model incorrectly classified 149 fraudulent claims as legitimate.

This is one of the most important metrics in fraud detection because false negatives represent missed fraudulent claims.

Compared with the baseline Decision Tree:

Baseline False Negatives: 137

Decision Tree + SMOTE False Negatives: 149

The SMOTE model actually produced more missed fraud cases.

This indicates that although SMOTE balanced the training data, it did not improve the Decision Tree's ability to generalize fraudulent patterns to unseen test data.

True Positives (TP) = 36

The model correctly identified 36 fraudulent claims.

Compared with the baseline Decision Tree:

Baseline True Positives: 48

Decision Tree + SMOTE True Positives: 36

The baseline Decision Tree detected more fraudulent cases than the SMOTE version.

This suggests that SMOTE did not provide additional benefit for this specific Decision Tree implementation and may have introduced synthetic patterns that did not fully represent real fraudulent claims.

Classification Report Analysis

Class 0: Non-Fraud Claims

Precision = 0.95

The model achieved 95% precision for legitimate claims.

This means that when the model predicts a claim as legitimate, it is correct approximately 95% of the time

The result is consistent with the baseline Decision Tree model, showing that SMOTE did not significantly impact the model's ability to classify non-fraud cases.

Recall = 0.93

The model correctly identified approximately 93% of legitimate claims.

Although slightly lower than the baseline Decision Tree:

Baseline Recall: 94%

SMOTE Recall: 93%

the performance remains strong.

F1-score = 0.94

The high F1-score indicates that the Decision Tree maintains excellent performance for the majority class.

Class 1: Fraud Claims

Precision = 0.15

The model achieved a fraud precision of 15%.

This means that only 15% of claims predicted as fraudulent were actually fraudulent.

Compared with the baseline Decision Tree:

Baseline Precision: 21%

SMOTE Precision: 15%

The precision decreased after applying SMOTE, meaning the model generated more false fraud predictions.

Recall = 0.19

The fraud recall was 19%.

This means that the model successfully detected 19% of all fraudulent claims.

Comparison:

Baseline Decision Tree Recall: 26%

SMOTE Decision Tree Recall: 19%

The baseline model performed better at identifying fraudulent claims.

F1-score = 0.17

The fraud F1-score decreased compared with the baseline:

Baseline Decision Tree F1-score: 0.23

SMOTE Decision Tree F1-score: 0.17

This indicates that the balance between fraud precision and recall became worse after applying SMOTE.

Comparison: Decision Tree Baseline vs Decision Tree with SMOTE

Metric	Decision Tree Baseline	Decision Tree + SMOTE
Accuracy	90%	88%
Fraud Precision	21%	15%
Fraud Recall	26%	19%
Fraud F1-score	23%	17%
True Fraud Detected	48	36
False Positives	176	212
False Negatives	137	149

Performance Comparison Analysis

1. Fraud Detection Capability

The baseline Decision Tree performed better at identifying fraudulent claims.

The number of detected fraud cases decreased:

Before SMOTE: 48 fraud cases detected

After SMOTE: 36 fraud cases detected

This indicates that SMOTE did not improve fraud recognition for this model.

2. Increased False Positives

The SMOTE-based model produced more false alarms:

Baseline: 176 false positives

SMOTE: 212 false positives

This means the model became more aggressive in identifying fraud but at the cost of incorrectly flagging legitimate claims.

3. Reduced Overall Performance

Unlike Logistic Regression, where SMOTE produced small improvements, the Decision Tree model experienced a decline in fraud detection performance.

Possible reasons include:

Synthetic Data Limitations

SMOTE creates artificial fraud cases by interpolating existing minority samples. These synthetic observations may not accurately represent real fraudulent behavior.

Decision Tree Sensitivity

Decision Trees can easily memorize training patterns. Introducing synthetic samples may cause the tree to learn artificial decision boundaries that do not generalize well to real test cases.

Overfitting Risk

Decision Trees are prone to overfitting, and additional synthetic observations may increase model complexity without improving generalization.

The Decision Tree model after applying SMOTE demonstrates:

Strengths:

- Maintains strong classification of legitimate claims
- Captures non-linear relationships between features

- Provides interpretable decision rules

Weaknesses:

- Lower fraud detection performance compared with baseline
- Increased false positives
- Lower fraud precision and recall
- SMOTE did not improve minority class detection

Applying SMOTE to the Decision Tree model did not improve fraud detection performance. While SMOTE successfully balanced the training data, the synthetic fraud samples did not appear to provide additional predictive value for this model.

The baseline Decision Tree actually achieved better fraud detection performance, suggesting that alternative approaches may be more effective.

Future improvements will focus on:

- Hyperparameter tuning (maximum depth, minimum samples split, pruning)
- Class-weighted Decision Trees
- Random Forest with SMOTE
- Gradient Boosting algorithms
- XGBoost
- Ensemble methods combining multiple classifiers

Tree-based ensemble models are expected to outperform a single Decision Tree because they reduce overfitting and capture more robust patterns associated with fraudulent insurance claims.

Conclusion

This project focused on developing and evaluating machine learning models for insurance fraud detection using historical insurance claim data. The primary objective was to identify patterns within customer, policy, vehicle, and claim-related attributes that could help distinguish fraudulent claims from legitimate insurance claims.

The project followed complete data analytics and machine learning workflow, including exploratory data analysis (EDA), data preprocessing, feature engineering, statistical feature analysis, model development, and performance evaluation. The dataset was analyzed to understand fraud patterns and identify the variables that contribute most significantly to fraudulent claim behavior.

During the exploratory analysis phase, several important relationships were identified between fraud occurrence and different claim characteristics. Features such as Fault, Base Policy, Vehicle Category, Policy Type, Past Number of Claims, Accident Area, Vehicle Price, and Agent Type demonstrated significant relationships with fraudulent claims. These findings indicate that fraud is influenced by a combination of policy-related, vehicle-related, and accident-related factors rather than a single individual attribute.

Data preprocessing was performed to prepare the dataset for machine learning algorithms. This included handling categorical variables through encoding techniques, removing irrelevant features such as Policy Number, and eliminating redundant attributes such as Age, since similar information was already represented by Age Of PolicyHolder. These preprocessing steps improved data quality and ensured compatibility with machine learning models.

Multiple classification models were developed and evaluated, including:

- Logistic Regression baseline model
- Decision Tree baseline model
- Logistic Regression with SMOTE
- Decision Tree with SMOTE

The Logistic Regression baseline model achieved a high overall accuracy of 94%, but its fraud detection capability was limited, identifying only a very small percentage of fraudulent claims. The model achieved excellent performance for legitimate claims but failed to capture minority-class fraud patterns due to the significant class imbalance in the dataset.

The Decision Tree baseline model demonstrated improved fraud detection capability compared with Logistic Regression. Although its overall accuracy was lower at 90%, it achieved better fraud recall by identifying a greater number of fraudulent claims. This demonstrated that tree-

based models were better suited for capturing the complex, non-linear relationships present in insurance fraud data.

To address the class imbalance problem, SMOTE was applied to generate synthetic fraudulent claim samples and provide the models with a more balanced training dataset. While SMOTE improved the representation of minority-class observations, the results varied across models. The Logistic Regression model showed a slight improvement in fraud detection performance after SMOTE, increasing fraud precision and recall. However, the improvement remained limited because Logistic Regression assumes linear relationships between variables.

The Decision Tree model with SMOTE did not outperform the baseline Decision Tree model. Although SMOTE increased the number of fraud examples during training, the model produced more false positives and achieved lower fraud detection metrics. This suggests that synthetic samples generated by SMOTE may not have fully represented real fraudulent behaviors, and additional optimization techniques are required.

Overall, the analysis demonstrated that fraud detection is a challenging classification problem due to severe class imbalance and complex relationships among variables. Accuracy alone was found to be an unreliable evaluation metric because a model can achieve high accuracy by simply predicting the majority class. Therefore, fraud-focused evaluation metrics such as precision, recall, F1-score, and ROC-AUC provide a more meaningful assessment of model performance.

The results highlight that machine learning models can provide valuable assistance in identifying suspicious insurance claims; however, further improvements are necessary before deployment in a real-world insurance environment.

Future Improvements

Although the developed models provide a strong foundation for insurance fraud detection, several improvements can be implemented to enhance predictive performance and practical applicability.

1. Implement Advanced Ensemble Machine Learning Models

Future work will explore more advanced algorithms capable of capturing complex relationships within insurance claim data.

Potential models include:

Random Forest

Random Forest combines multiple decision trees to improve prediction stability and reduce overfitting. By averaging predictions from multiple trees, it can capture diverse fraud patterns more effectively than a single Decision Tree.

Expected benefits:

- Improved generalization
- Higher fraud detection accuracy
- Reduced overfitting
- Better feature importance analysis
- Gradient Boosting Models

Models such as:

- Gradient Boosting Classifier
- XGBoost
- LightGBM
- CatBoost

will be evaluated because they are widely used in fraud detection applications.

These models iteratively improve weak learners and can identify subtle relationships between policy characteristics and fraudulent behavior.

Expected improvements:

- Higher fraud recall
- Better F1-score
- Improved ROC-AUC performance

2. Hyperparameter Optimization

The baseline models were developed using default parameters. Future improvements should include systematic hyperparameter tuning using techniques such as:

- Grid Search
- Random Search
- Bayesian Optimization

Important parameters to optimize include:

Decision Tree:

Maximum tree depth

Minimum samples per split

Minimum samples per leaf

Splitting criterion

Logistic Regression:

Regularization strength (C parameter)

Solver selection

Class weighting

Hyperparameter tuning can improve model performance while reducing overfitting.

3. Experiment with Alternative Sampling Techniques

Although SMOTE was evaluated, other imbalance handling techniques should also be considered.

Possible approaches include:

- ADASYN (Adaptive Synthetic Sampling)

Creates more synthetic samples in areas where fraud cases are difficult to classify.

- Random Undersampling

Reduces the majority class size to create a more balanced dataset.

- SMOTE + Tomek Links

Combines oversampling with removal of overlapping samples to improve class separation.

Comparing multiple sampling strategies may identify the best approach for this dataset.

4. Feature Engineering Improvements

Additional feature engineering could improve fraud detection performance.

Potential engineered features include:

Claim Risk Score

Combining multiple fraud indicators:

- No police report
- No witnesses
- Previous claim history
- External agent involvement
- Vehicle Risk Indicators

Creating features based on:

- Vehicle age
- Vehicle price
- Vehicle category
- Repair cost estimates

Policyholder Behavior Features

Examples:

- Number of previous claims per year
- Time between policy purchase and claim submission
- Frequency of policy changes

These engineered variables may provide stronger fraud signals than individual attributes.

Final Summary

The insurance fraud detection project successfully demonstrated the ability of machine learning techniques to identify patterns associated with fraudulent claims. The analysis showed that while traditional models such as Logistic Regression provide strong baseline performance, they struggle with the complexity and imbalance of fraud data. Decision Tree models improved fraud detection by capturing non-linear relationships, but additional optimization is required.

Future improvements should focus on advanced ensemble algorithms, improved sampling strategies, feature engineering, cost-sensitive learning, hyperparameter optimization, and explainable AI techniques. These enhancements would create a more accurate, reliable, and practical fraud detection system capable of supporting insurance companies in reducing fraudulent claim losses while maintaining efficient claim processing.

Responsible AI Productivity (Coding-Assist Declaration and Verification)

Coding-Assist Declaration

This project utilized AI-assisted coding tools as a productivity enhancement resource during the development of the insurance fraud analytics workflow. AI assistance was used to support programming tasks such as generating code templates, debugging errors, improving code readability, suggesting data analysis approaches, and providing explanations of machine learning concepts.

The use of AI tools was limited to assisting with implementation and learning. All analytical decisions, data preprocessing approaches, feature selection choices, model development strategies, evaluation methods, and interpretation of results were reviewed and validated by the project author.

AI-generated suggestions were not directly accepted without verification. Any code, methodology, or explanations generated through AI assistance were evaluated, modified where necessary, and tested within the project environment to ensure correctness and alignment with the project objectives.

Responsible AI Considerations

The use of AI assistance followed responsible AI practices by ensuring:

Transparency

AI assistance was openly documented, and its role was limited to supporting productivity and development.

Human Oversight

All final decisions regarding:

Data preprocessing

Feature selection

Model selection

Performance interpretation

were made and verified by myself.

Accuracy and Reliability

AI-generated code and recommendations were tested before being incorporated into the project.

Ethical Use

AI tools were not used to replace critical thinking, automate decision-making without validation, or generate unsupported conclusions.

Final Declaration

I confirm that AI-assisted tools were used responsibly as a coding and learning support resource throughout this project. All AI-generated suggestions were reviewed, tested, and validated before use. The final implementation, analysis, conclusions, and project outcomes represent my own understanding, evaluation, and interpretation of the insurance fraud analytics problem.